

Brain-predictivity as the honest test: the Cichy-92 benchmark for the Gömb–Soma vision model

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Abstract

The Gömb–Soma vision stack — a signed-hypergraph object model built on an adaptive quadtree, Forman–Ricci curvature, Hodge decomposition and curvature-driven rewiring — does *not* win at small-scale image classification: in a fair re-benchmark it trails even a plain MLP on MNIST and Fashion-MNIST. That is a falsification of the *accuracy* hypothesis, not of the model’s reason to exist. This note argues that the right test for a structurally-grounded, brain-inspired vision model is not classification accuracy but *brain-predictivity*: how well its internal representations explain measured cortical activity. We describe the Cichy 92-image MEG/fMRI benchmark [1], the representational-similarity and Brain-Score methodology [2, 4], the parameter-matched protocol already implemented in this repository, and the single missing piece — the real fMRI data — needed to run it.

1 Why this benchmark, and why now

A vision model can be judged on two very different axes. The first is *task performance* (does it classify/detect well?). On this axis the Gömb–Soma family is, honestly, a negative result at the scales we can afford: the 2026-05-28 re-benchmark (CNN, MLP, HSiKAN, NeocogHGNN, RicciStim on MNIST/Fashion-MNIST, 15 epochs $\times$ 3 seeds) places RicciStim and the hypergraph operators *below* a vanilla MLP, and the differential-geometric machinery does not recover the gap. The structural prior, on natural-image classification at small scale, hurts.

The second axis is *representational alignment with biological vision*: a model is interesting to neuroscience if its layer activations linearly predict the responses of visual cortex, *independently* of whether it tops a classification leaderboard. Many high-accuracy networks are poor cortical predictors, and some lower-accuracy models align surprisingly well; accuracy and brain-predictivity are related but distinct [3, 4]. Since Gömb–Soma is explicitly motivated by cortical structure — coarse-to-fine receptive fields, signed lateral interactions, curvature/flow on a retinotopic graph — the fair question is the second one. The Cichy-92 benchmark is the standard, compact instrument for asking it.

2 The Cichy 92-image dataset

Cichy, Pantazis and Oliva [1] presented 92 isolated object images (human bodies, faces, animals, natural and artificial objects) to human participants while recording **both** MEG (millisecond temporal resolution) and fMRI (millimetre spatial resolution). The two modalities are fused through *representational similarity analysis* (RSA) [2]: each modality, region and time point yields a 92×92 *representational dissimilarity matrix* (RDM) over the image pairs, and models are compared to brains in this common RDM space rather than in raw activations. For the spatial (fMRI) side the canonical regions of interest are early visual cortex (V1) through higher

object-selective cortex (e.g. IT/LOC), which makes the dataset a natural probe for a *hierarchical* model: a coarse-to-fine feature stack should map onto the early-to-late cortical gradient.

The set is small (92 conditions), which is a feature: it is large enough for stable RDM estimation with split-half noise ceilings, and small enough that a parameter-light model and a parameter-matched baseline can be compared end-to-end in seconds-to-minutes per seed.

3 Scoring methodology (Brain-Score protocol)

We follow the Brain-Score pipeline [4], implemented here without the external `brainscore` package (no network dependency):

1. **Feature extraction.** For each of the 92 images, take a model’s per-stage activations and bin them retinotopically (per quadtree depth for Gömb–Soma; per residual block for the ResNet baseline).
2. **Dimensionality reduction.** PLS regression compresses model features toward the neural target, guarding against the high-dimensional-features / few-conditions regime.
3. **Per-voxel / per-ROI ridge mapping.** A cross-validated linear (ridge) map from model features to each ROI’s responses; the score is the held-out predictivity (Pearson r or R^2).
4. **Noise-ceiling correction.** Split-half reliability of the neural data bounds the achievable score; reported predictivity is normalised by this ceiling so the number is interpretable as “fraction of explainable variance captured”.

The comparison is *parameter-matched*: a ~ 30 k-parameter ResNet-tiny is scored against the ~ 30 k-parameter Gömb–Soma backbone on the identical pipeline, so any difference reflects representational geometry rather than capacity.

4 The Gömb–Soma cortical hypothesis

The model offers three concrete, cortically-motivated features whose brain-predictivity is testable:

- **Per-depth anchors as a V1→V4 gradient.** The adaptive quadtree subdivides the image where salience/variance is high; shallow depths are coarse and uniform (V1-like), deep depths are fine and content-driven (V4/IT-like). Per-depth binned features are the model’s “cortical stages”.
- **Forman–Ricci curvature** [5] on the patch graph as a boundary/edge descriptor — a candidate correlate of contour and figure-ground signals in early/mid visual areas.
- **Hodge eigenmodes** of the patch-graph Laplacian as global shape descriptors, and signed lateral edges (the σ structure) as a model of excitatory/inhibitory lateral interaction.

The hypothesis is falsifiable and sharp: *at matched parameters, the Gömb–Soma per-depth features predict V1/V2/V4 fMRI RDMs at least as well as a ResNet-tiny*, with a paired bootstrap over the 92 conditions giving the significance. A negative result here — no better than ResNet — would retire the cortical-prototype framing; a positive result would be the model’s first genuine win, on the axis it was actually designed for.

5 Implementation status and what is missing

The scoring pipeline is built and tested in `signedkan_wip/src/cortical/`: `CorticalFeatureExtractor` (per-depth retinotopic binning), `BrainScorer` (PLS → ridge → k -fold CV → split-half noise ceiling), a parameter-matched `ResNetTinyCortical` baseline, and a shape-faithful *synthetic* Cichy-92 generator that drives an end-to-end smoke (92 images $\times$ 16 subjects in ~ 22 s/seed, 21 unit tests). The API is deliberately dataset-agnostic: ingesting the real data is “one new adapter function”.

The one missing piece is the **real Cichy-92 fMRI data**. The synthetic generator validates shapes and the pipeline, but *cannot* test the hypothesis: whichever model scores higher on

generated RDMS is an artefact of the generator. Running the real benchmark requires (i) the published Cichy-92 fMRI RDMS (or beta maps) in a known format, and (ii) the ~ 50 -line adapter that maps them onto the existing **BrainScorer** interface. Both are small once the data source is fixed.

6 Conclusion

Gömb-Soma should not be defended as a classifier — the evidence says it is not one. Its honest evaluation is brain-predictivity, and Cichy-92 is the compact, standard instrument for it. The protocol, baseline and code are ready; the next concrete step is acquiring the real fMRI RDMS and writing the adapter, after which a single parameter-matched run settles whether the cortical-prototype framing survives.

References

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